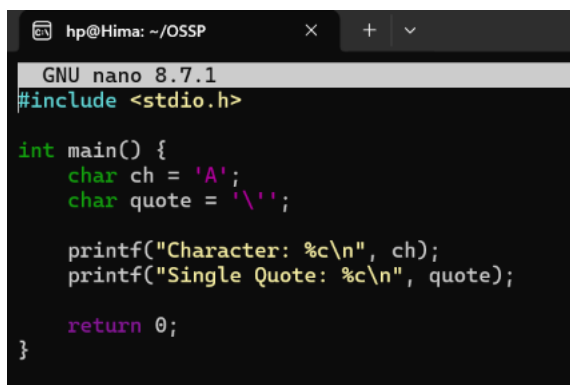


5. To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

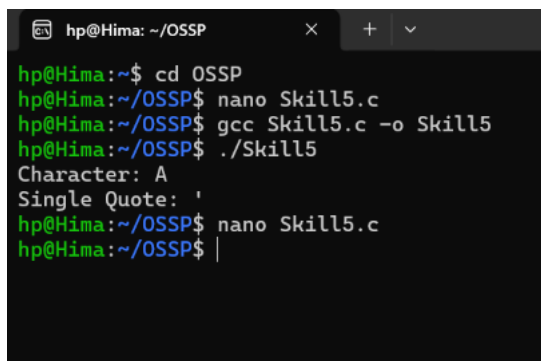


```
hp@Hima: ~/OSSP
GNU nano 8.7.1
#include <stdio.h>

int main() {
    char ch = 'A';
    char quote = '\'';

    printf("Character: %c\n", ch);
    printf("Single Quote: %c\n", quote);

    return 0;
}
```



```
hp@Hima: ~/OSSP
hp@Hima:~$ cd OSSP
hp@Hima:~/OSSP$ nano Skill5.c
hp@Hima:~/OSSP$ gcc Skill5.c -o Skill5
hp@Hima:~/OSSP$ ./Skill5
Character: A
Single Quote: '
hp@Hima:~/OSSP$ nano Skill5.c
hp@Hima:~/OSSP$ |
```

6. To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

```
hp@Hima: ~/OSSP
GNU nano 8.7.1

printf("Enter a string with escape sequences:\n");
fgets(input, sizeof(input), stdin);

for(i = 0; input[i] != '\0'; i++) {
    if(input[i] == '\\') {
        i++;
        switch(input[i]) {
            case 'n':
                output[j++] = '\n';
                break;
            case 't':
                output[j++] = '\t';
                break;
            case '\\':
                output[j++] = '\\';
                break;
            case ' ':
                output[j++] = ' ';
                break;
            case '*':
                output[j++] = '*';
                break;
            case '\':
                output[j++] = '\';
                break;
            default:
                output[j++] = '\\';
                output[j++] = input[i];
        }
    } else {
        output[j++] = input[i];
    }
}

output[j] = '\0';

printf("\n--- Parser Output ---\n");
printf("%s\n", output);

printf("\nLength: %lu\n", strlen(output));

return 0;
}
```

```
hp@Hima:~/OSSP$ nano Skill6.c
hp@Hima:~/OSSP$ gcc Skill6.c -o Skill6
hp@Hima:~/OSSP$ ./Skill6
Enter a string with escape sequences:
H i m a

--- Parser Output ---
H i m a

Length: 8
hp@Hima:~/OSSP$ |
```

7. Process Synchronization and Child Process Monitoring Using waitpid().

To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution.

```
hp@Hima: ~/OSSP
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid;
    int status;

    pid = fork();

    if (pid < 0) {
        printf("Fork failed.\n");
        return 1;
    }

    if (pid == 0) {
        printf("Child Process Started\n");
        printf("Child PID: %d\n", getpid());

        sleep(3);

        printf("Child Process Finished\n");
        exit(25);
    }
    else {
        printf("Parent PID: %d\n", getpid());
        printf("Waiting for Child Process...\n");

        waitpid(pid, &status, 0);

        if (WIFEXITED(status)) {
            printf("Child exited normally.\n");
            printf("Exit Status: %d\n", WEXITSTATUS(status));
        }
        else {
            printf("Child did not exit normally.\n");
        }
    }

    return 0;
}
```

```
hp@Hima:~/OSSP$ nano skill7.c
hp@Hima:~/OSSP$ gcc skill7.c -o skill7
hp@Hima:~/OSSP$ ./skill7
Parent PID: 680
Waiting for Child Process...
Child Process Started
Child PID: 681
Child Process Finished
Child exited normally.
Exit Status: 25
hp@Hima:~/OSSP$ |
```

```
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr
/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorp
orationII.WindowsSubsystemForLinux.2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/Progra
m Files (x86)/VMware/VMware Workstation/bin:/mnt/c/Program Files/Common Fil
es/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/
java8path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/
c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/window
s/System32/WindowsPowerShell/v1.0:/mnt/c/windows/System32/OpenSSH:/mnt/c/P
rogram Files (x86)/NVIDIA Corporation/PhysX/Common:/mnt/c/Program Files/NVID
IA Corporation/NVIDIA NvDLISR:/mnt/c/Users/Administrator/AppData/Local/Micro
soft/WindowsApps:/mnt/c/Program Files/HP/OMEN-Broadcast/Common:/mnt/c/Progra
m Files/nodejs:/mnt/c/ProgramData/chocolatey/bin:/mnt/c/Program Files/Git/c
md:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/reduk/AppDat
a/Local/Microsoft/WindowsApps:/mnt/c/Users/reduk/AppData/Local/Programs/Micr
osoft VS Code/bin:/mnt/c/Program Files/JetBrains/IntelliJ IDEA 2025.2.5/bin:
```

8. To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Logic.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

```
hp@Hima: ~/OSSP
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

int main() {
    char input[200], output[500];
    int i = 0, j = 0;

    printf("Enter a string: ");
    fgets(input, sizeof(input), stdin);

    while (input[i] != '\0') {
        if (input[i] == '$') {
            i++;
            char var[100];
            int k = 0;

            while (isalnum(input[i]) || input[i] == '_') {
                var[k++] = input[i++];
            }
            var[k] = '\0';

            char *value = getenv(var);

            if (value != NULL) {
                strcpy(&output[j], value);
                j += strlen(value);
            } else {
                strcpy(&output[j], "[UNDEFINED]");
                j += strlen("[UNDEFINED]");
            }
        } else {
            output[j++] = input[i++];
        }
    }

    output[j] = '\0';

    printf("\nExpanded Output:\n%s", output);

    return 0;
}
```

```
hp@Hima:~/OSSP$ nano skill8.c
hp@Hima:~/OSSP$ gcc skill8.c -o skill8
hp@Hima:~/OSSP$ ./skill8
Enter a string: Himavarshini

Expanded Output:
Himavarshini
hp@Hima:~/OSSP$
```

9. To Change Directories, Validate Paths, Update Working Directory, Handle Errors, Maintain Previous Directory, Test Navigation.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

```
hp@Hima: ~/OSSP
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <limits.h>

int main() {
    char path[PATH_MAX];
    char current[PATH_MAX];
    char previous[PATH_MAX] = "";

    while (1) {
        getcwd(current, sizeof(current));
        printf("\nCurrent Directory: %s\n", current);

        printf("Enter directory (type exit to quit): ");
        fgets(path, sizeof(path), stdin);
        path[strcspn(path, "\n")] = '\0';

        if (strcmp(path, "exit") == 0)
            break;

        if (strcmp(path, "--") == 0) {
            if (strlen(previous) == 0) {
                printf("No previous directory.\n");
                continue;
            }
            strcpy(path, previous);
        }

        strcpy(previous, current);

        if (chdir(path) == 0) {
            getcwd(current, sizeof(current));
            printf("Directory Changed Successfully.\n");
            printf("New Directory: %s\n", current);
        } else {
            perror("Error");
        }
    }

    return 0;
}
```

```
hp@Hima:~/OSSP$ nano skill9.c
hp@Hima:~/OSSP$ gcc skill9.c -o skill9
hp@Hima:~/OSSP$ ./skill9

Current Directory: /home/hp/OSSP
Enter directory (type exit to quit): /home/hp
Error: No such file or directory

Current Directory: /home/hp/OSSP
Enter directory (type exit to quit): exit
hp@Hima:~/OSSP$
```

10. To Retrieve Current Directory, Display Path, Process Exit Requests, Cleanup Resources, Save State, Verify Correct Behavior.

To Parse Export Syntax, Update Environment Variables, Validate Names, Handle Existing Variables, Export to Child Processes, Test Functionality

```
hp@Hima: ~/OSSP
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
#include <unistd.h>
#include <limits.h>
#include <sys/wait.h>

int validName(char *name)
{
    if (!(isalpha(name[0]) || name[0] == '_'))
        return 0;

    for (int i = 1; name[i]; i++)
    {
        if (!(isalnum(name[i]) || name[i] == '_'))
            return 0;
    }
    return 1;
}

int main()
{
    char input[200];
    char cwd[PATH_MAX];
    char lastPath[PATH_MAX];

    while (1)
    {
        if (getcwd(cwd, sizeof(cwd)) != NULL)
        {
            strcpy(lastPath, cwd);
            printf("\nCurrent Directory: %s\n", cwd);
        }
        else
        {
            perror("getcwd");
            break;
        }

        printf("\nCommands:\n");
        printf("  pwd\n");
        printf("  export NAME=value\n");
        printf("  child\n");
        printf("  exit\n");
    }
}
```

```
hp@Hima: ~/OSSP
GNU nano 8.7.1 skill10.c *
printf("Exiting program...\n");
break;
}

/* Child Process */
else if (strcmp(input, "child") == 0)
{
    pid_t pid = fork();

    if (pid == 0)
    {
        execl("/usr/bin/env", "env", NULL);
        perror("exec");
        exit(1);
    }

    wait(NULL);
}

/* Export Command */
else if (strncmp(input, "export ", 7) == 0)
{
    char *expr = input + 7;
    char *eq = strchr(expr, '=');

    if (eq == NULL)
    {
        printf("Invalid syntax. Use: export NAME=value\n");
        continue;
    }

    *eq = '\0';
    char *name = expr;
    char *value = eq + 1;

    if (!validName(name))
    {
        printf("Invalid variable name.\n");
        continue;
    }

    setenv(name, value, 1);
    printf("Exported: %s=%s\n", name, getenv(name));
}

Help      Write Out  Where Is   Cut         Execute
Exit      Read File   Replace    Paste       Justify
Location  Go To Line  Undo      Redo
M-A      M-G      M-6
```

```
Current Directory: /home/hp/OSSP
Enter directory (type exit to quit): exit
hp@Hima:~/OSSP$ nano skill10.c
hp@Hima:~/OSSP$ gcc skill10.c -o skill10
hp@Hima:~/OSSP$ ./skill10

Current Directory: /home/hp/OSSP

Commands:
pwd
export NAME=value
child
exit

Enter command: pwd
Path: /home/hp/OSSP

Current Directory: /home/hp/OSSP

Commands:
pwd
export NAME=value
child
exit

Enter command: export
Invalid syntax. Use: export NAME=value

Current Directory: /home/hp/OSSP

Commands:
pwd
export NAME=value
child
exit
```

```
Commands:
  pwd
  export NAME=value
  child
  exit

Enter command: child
SHELL=/bin/bash
WSL_DISTRO_NAME=Ubuntu
NAME=Hima
PWD=/home/hp/OSSP
LOCALNAME=hp
HOME=/home/hp
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/326_interop
LS_COLORS=ls=0;ld=0;1;34;ln=0;1;36;mh=0;pi=0;33;se=0;1;35;sd=0;1;35;bd=0;33;01;cd=0;33;01;or=0;31;01;nd=0;35;su=0;37;41;sg=0;33;43;ca=0;30;tw=0;42;ow=0;42;st=0;37;44;ex=0;1;32;*tar=0;1;31;*tgs=0;1;31;*arc=0;1;31;*ar
j=0;1;31;*taz=0;1;31;*lha=0;1;31;*lzh=0;1;31;*lzm=0;1;31;*tlz=0;1;31;*txz=0;1;31;*tzo=0;1;31;*zip=0;1;31;*z=0;1;31;*dz=0;1;31;*gz=0;1;31;*lrx=0;1;31;*lz=0;1;31;*lzo=0;1;31;*xz=0;1;31;*
.zst=0;1;31;*tzt=0;1;31;*bz2=0;1;31;*bz=0;1;31;*tbz=0;1;31;*tbz2=0;1;31;*tz=0;1;31;*deb=0;1;31;*rpm=0;1;31;*jar=0;1;31;*war=0;1;31;*ear=0;1;31;*sar=0;1;31;*rar=0;1;31;*alz=0;1;31;*ace=0;1;31;*zoo=0;1;31;*cpi
o=0;1;31;*7z=0;1;31;*rz=0;1;31;*cab=0;1;31;*wim=0;1;31;*swm=0;1;31;*dwm=0;1;31;*esd=0;1;31;*avif=0;1;35;*jpg=0;1;35;*jpeg=0;1;35;*njpg=0;1;35;*gif=0;1;35;*bmp=0;1;35;*pbm=0;1;35;*pgm=0;1;35;*ppm
=0;1;35;*tga=0;1;35;*xbm=0;1;35;*xpm=0;1;35;*tif=0;1;35;*tiff=0;1;35;*png=0;1;35;*svg=0;1;35;*svgz=0;1;35;*mng=0;1;35;*pc=0;1;35;*mov=0;1;35;*mpeg=0;1;35;*m2v=0;1;35;*m4v=0;1;35;*webm=0;1;35;*web
p=0;1;35;*ogg=0;1;35;*mpu=0;1;35;*m4u=0;1;35;*mp4v=0;1;35;*vob=0;1;35;*qt=0;1;35;*nuv=0;1;35;*wmv=0;1;35;*asf=0;1;35;*rm=0;1;35;*rmvb=0;1;35;*flc=0;1;35;*avi=0;1;35;*fli=0;1;35;*flv=0;1;35;*gl=0;1;35;*dl=0;1;3
5;*xcf=0;1;35;*xwd=0;1;35;*yuv=0;1;35;*cgm=0;1;35;*emf=0;1;35;*ogv=0;1;35;*ogx=0;1;35;*aac=0;0;36;*au=0;0;36;*flac=0;0;36;*m4a=0;0;36;*mid=0;0;36;*midi=0;0;36;*mka=0;0;36;*mp3=0;0;36;*mpc=0;0;36;*ogg=0;0;36;*
.ra=0;0;36;*wav=0;0;36;*oga=0;0;36;*opus=0;0;36;*spx=0;0;36;*xspf=0;0;36;*#*=0;0;90;*#*=0;0;90;*bak=0;0;90;*old=0;0;90;*orig=0;0;90;*part=0;0;90;*rej=0;0;90;*swp=0;0;90;*tmp=0;0;90;*dpg=0;0;90;*dpg-dist=0;0;90;*dpg-old=0;
90;*ucf-dist=0;0;90;*ucf-new=0;0;90;*ucf-old=0;0;90;*rpmnew=0;0;90;*rpmorig=0;0;90;*rpmsave=0;0;90;
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=hp
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/1000
WSLENv=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/napd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/usr/games:/usr/local/games:/usr/lib/WSL/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII/WindowsSubsystemForLinux_2.7.11.0_x64__8w
4ky9d8bbwe:/mnt/c/Program Files (x86)/Nmax/Where Where Notation/bin:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files
(x86)/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program File
s/nodejs:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/HP/Downloads/apache-maven-3.9.16-bin:/mnt/c/Users/HP/AppData/Local/Microsoft/WindowsApps/mnt/c/Users/HP/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c
/Users/HP/AppData/Roaming/npm:/mnt/c/Users/HP/AppData/Local/Kingsoft/MS Office/12.1.0.27458/clitool:/mnt/c/Users/HP/AppData/Local/Python/bin:/mnt/c/apache-maven-3.9.16-bin/apache-maven-3.9.16/bin:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
PULSE_SERVER=unix:/mnt/WSL/PulseServer
OLDPWD=/home/hp
././skill10

Current Directory: /home/hp/OSSP
```

```
OLDPWD=/home/hp
././skill10

Current Directory: /home/hp/OSSP

Commands:
  pwd
  export NAME=value
  child
  exit

Enter command: exist
Invalid command.

Current Directory: /home/hp/OSSP

Commands:
  pwd
  export NAME=value
  child
  exit

Enter command: exit

Saving last directory: /home/hp/OSSP
Exiting program...
hp@Hima:~/OSSP$ |
```